Probing the limit of forward materials screening and the novelty claimed by inverse design with 14.6 million crystals resolved across 55 databases

Zhilong Song[1,2†], Zijun Pang[4†], Lixue Cheng[1,2,3*]

[1]Department of Chemistry, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.
[2]IAS Center for AI for Scientific Discoveries, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.
[3]Department of Chemical and Biological Engineering, Hong Kong University of Science and Technology, Kowloon, Hong Kong 999077, China.
[4]School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, Sichuan 611731, China.

*Corresponding author. E-mail: lixuecheng@ust.hk
†These authors contributed equally to this work.

Abstract

Materials discovery follows two routes. Forward screening searches the crystals already known for one with the property wanted. Inverse design generates a crystal to order and reports how many of its products are new. Both depend on a question no public database can answer alone: is this crystal already known, and is this property already known for it? The databases share no key at the level of the crystal, so the same structure carries a different identifier in each. Here we resolve that identity across 55 public databases and keep, for every value, the source and the method behind it. The result holds 14,590,485 distinct crystals and 131.8 million values over 166 properties, and it grows as it ingests, because a record matching nothing becomes the next member of the reference. For forward screening, the record locates the limit. It is not the number of known crystals but two things a campaign inherits from whichever database it loads. Of 846,303 crystals carrying a formation energy from more than one source at one level of theory, only 65.3% agree within 50 meV atom^{-1}, and the disagreement is a systematic offset rather than noise. Of fifty properties needing a calculation of their own, none has been computed for a tenth of the pool. For inverse design, the record is the reference against which novelty has to be judged, and recent campaigns have reported as new what such a reference already held. The corpus supplies 513 bounded learning tasks to complete the property layer over the whole record, and one identity criterion under which novelty can be stated.

Keywords: materials databases, data integration, crystal structure deduplication, provenance, cross-database benchmarking, machine learning for materials

1 Introduction

Materials discovery follows two routes. Forward screening searches the crystals already known for one that has the property wanted. Inverse design generates a crystal to order, and reports as its main figure of merit how many of its products are new. Each route is usually described as bounded by something of its own: forward screening by how many crystals are known, inverse design by how good the generator is. Neither bound is where it is placed. Both routes rest on one question, asked of every crystal they touch: is this structure already known, and is this property already known for it? No public database can answer that question on its own.

The reason is that the databases share no key at the level of the crystal. Public repositories hold millions of computed structures [1–8], and generative models and universal interatomic potentials are trained on them [9–16]. But the same arrangement of atoms carries a different accession number in every database that holds it, and the chemical formula cannot serve as a key because polymorphs share one. So the union cannot be deduplicated, values from different providers cannot be placed on the same object, and three things stay unmeasured: how much of the record is one crystal counted several times, whether two sources that computed the same quantity agree, and which properties are known for which crystals. OPTIMADE gave the databases a common query interface [17, 18] and LeMaterial published a deduplicated union of several large ones [19]. Neither delivers the property layer, because two databases publishing a column called band gap are often not reporting the same physical quantity, and a union that merges by column name averages a semilocal, a hybrid and a measured value into one number.

That absence bounds each route in a specific way. A forward screen can rank crystals that nobody has computed, since that is what a surrogate model is for. What it cannot do is reach beyond the pool it was given or trust its labels more than the sources that produced them. A campaign screens whichever database it loaded and trains on that database's values, so the limit of forward screening is set by the size of the pool and the consistency of the labels, and both come from the record. Inverse design reports novelty against a reference assembled from a few of the largest repositories [19, 20], so the limit of what it may call new is set by the reference and by the criterion used to match against it, and both again come from the record. Three recent assessments make the point. A survey of one campaign's 384,870 stable predictions found them to be modifications of known materials [21]. The flagship compound of a generative model was later identified as a disordered phase reported in 1972 and present in the model's own training set [22]. And an ion-exchange

baseline built from known compounds outperformed four generative techniques [23]. In each case the record already contained what was reported as new.

Here we build the record that both routes need. Identity is resolved across 55 databases at the level of the crystal structure. Every value keeps the source, the native record and the method that produced it, so that several sources speaking about one crystal stay separately audible. Licence boundaries are enforced in the pipeline. Admission and recognition are one operation: a record that matches nothing becomes the corpus's next member, and its fingerprint joins the criterion applied to every record after it, so the corpus grows as it ingests rather than being rebuilt. The Results first describe the record, then what a forward screen inherits from it, then what a completion of it would settle, and finally what it means for a claim of novelty.

2 Results

2.1 One record from 55 databases

Making 55 databases comparable comes down to one decision, taken once per incoming record: is this crystal already in the corpus? We settle it on the structure rather than on the label. Each record is matched first on a cross-database identifier the source is entitled to assert, then on a structural fingerprint, and never on the chemical formula, which polymorphs share (Fig. 1a; Methods). A record that matches nothing is admitted rather than discarded, and every value it carries keeps the source, the native record and the method behind it. That second choice has a consequence this section measures: because an admitted record joins the criterion applied to the next, what results is not a merge of 55 databases but a corpus that extends the substrate it recognises them against.

The corpus so assembled holds 14,590,485 distinct crystal structures, of which 357,127 are experimental and 14,233,358 are computational candidates. On that pool sit 131,824,881 property values spanning 166 properties, 1,198,897 stored spectra and 21,543,633 references recording which database contains which crystal.

The contributors are extremely unequal, spanning 5.7 orders of magnitude, with the largest supplying 46.0% of all values and 30 of the 55 supplying fewer than 10^5 each (Fig. 1b). The two ends of that distribution do different work: the large general-purpose databases determine how much of the pool carries any property at all, while the small specialised libraries determine how many distinct properties exist to carry [24–26] — the most property-diverse contributor supplies 61 of the 166 properties, the largest supplies 10.

The corpus is considerably larger than the pool it started from, and the difference is exactly the records that matched nothing (Fig. 1c). The deduplicated starting pool held 8,225,138 structures, and admission added 6,365,347 more, a growth of 77.4%. Of the 21,543,633 references, 6,784,564 were resolved by record identifier, 8,294,560 by structural fingerprint alone, and the remainder record a database's own entry (Fig. 1d). The two matched routes are not interchangeable: identifier matching is exact but available only where a database publishes a cross-reference it owns, while fingerprint matching is available everywhere but has a domain of validity.

That domain has an edge, and it is worth stating before any conclusion is drawn from the corpus. Enumerated framework libraries are generated by decorating a small number of parent topologies, so sibling entries are identical in everything a bonding-graph fingerprint reads and more than 10^5 distinct hypothetical frameworks would collapse onto a few hundred pool materials. Such libraries are detected at the level of the source rather than the record and admitted under a composite key, and the boundary of validity this establishes is stated in Supplementary Note S2.

A structure admitted from one database becomes recognisable to every database that follows, because its fingerprint and identifiers join the matching substrate as it is admitted. The corpus therefore does not merely accumulate: the criterion by which it recognises the next record is itself a product of the records already ingested, and it is by this mechanism that the database keeps itself current as sources publish new versions and new sources appear. The effect is measurable. Of the 15,079,124 references resolved by matching, 3,676,707, or 24.4%, land on a crystal that was absent from the starting pool and was admitted during assembly (Fig. 2c). Those links do not exist in a construction that matches every database against a fixed reference. The figure is a property of the assembly rather than of the sources, since a different ingestion order would distribute it differently between them, and it is a lower bound on what a frozen reference would have missed rather than a measure of quality. What it establishes is that such links exist at all, and how much of the record they reach.

The effect is ordered by ingestion time and by chemical distance from the starting pool. AFLOW, ingested about ninety minutes after Alexandria contributed 1,514,488 structures, resolved 47.7% of its 1,401,895 matches onto crystals admitted during assembly. The JARVIS tight-binding set, whose fitting geometries are deliberately non-equilibrium and largely absent from the starting pool, reached 67.6%. The framework libraries show the effect in its purest form: CoRE MOF, ingested last among them, resolved 99.5% of its matches onto frameworks contributed by four earlier libraries. NOMAD [7], the last source ingested of the 55 and so the one matched against the largest substrate, resolved

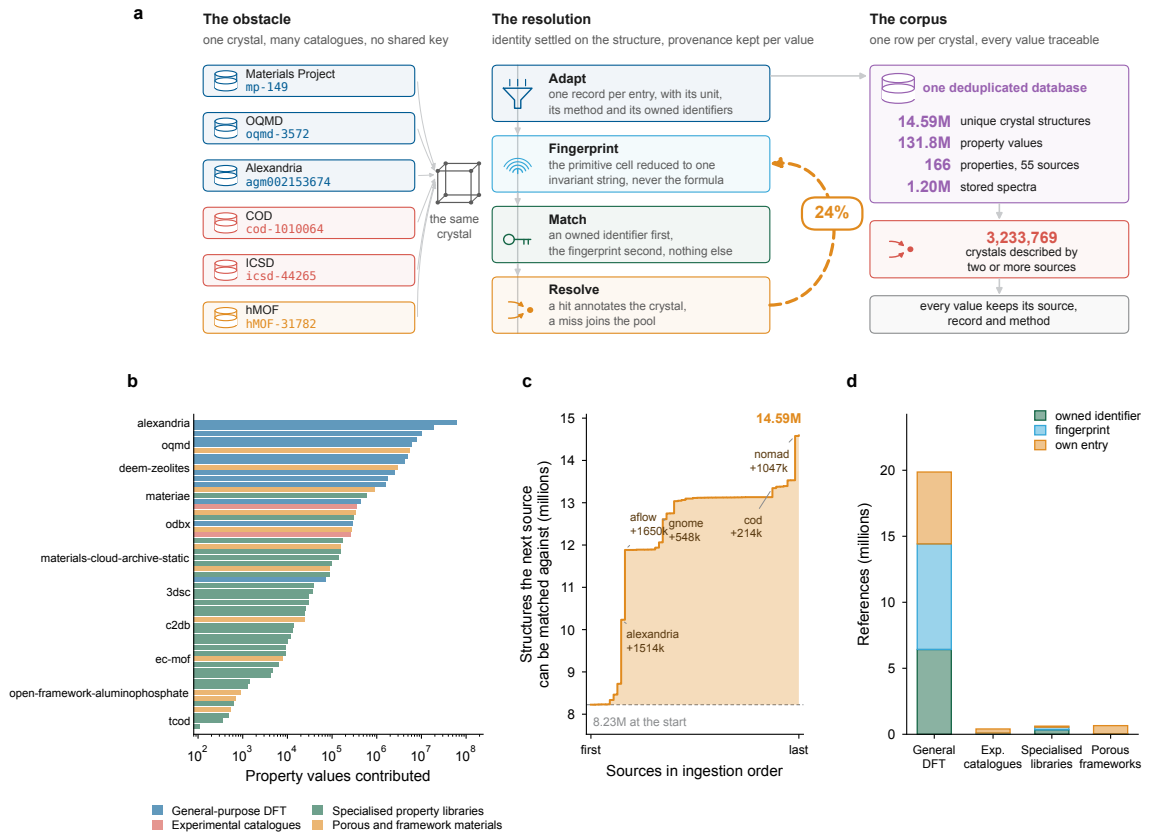


Fig. 1 Making the source databases comparable at the level of the crystal. **a**, The obstacle and the resolution. Each database labels the same arrangement of atoms with its own accession number, and no key relates them. Every incoming record is reduced to a structural fingerprint and matched first on a cross-database identifier the source is entitled to assert, second on that fingerprint, never on the chemical formula. The dashed loop is the mechanism that makes the corpus self-evolving: a record matching nothing is admitted, and its fingerprint re-enters the matcher, so each source is matched against a larger pool than the one before it. The figure on the loop is the percentage of all matches that land on a crystal admitted this way. The right column reports what the assembled corpus holds. **b**, Property values contributed by each database on a logarithmic axis, coloured by family, with labels at regular intervals through the ranking. **c**, The loop of **a**, measured. Each step is one source's admissions entering the match map, so the height at any point is the number of crystals the next source could be matched against. The largest single contributions are labelled. **d**, How the references were resolved, grouped by database family: by an identifier the source owns, by structural fingerprint, or as a record that the database contains the structure itself.

56.1% of its 3,168,530 matches onto crystals the starting pool did not contain, and contributed 1,047,208 structures of its own that now form part of the criterion applied to whatever is ingested next. Databases whose chemistry the starting pool already covered show the converse, at or near zero.

The practical consequence is that the marginal value of adding a database is not confined to the values it brings. CoRE MOF contributed 155,488 experimental porosity measurements, and the reason 99.5% of them landed on an existing crystal rather than forming an isolated set of new ones is that earlier libraries had already put those frameworks in the pool. Assembled against a frozen reference, the same measurements would have arrived as an unconnected island, and the corpus would have to be rebuilt from the beginning to connect them.

2.2 What a forward screen inherits

A forward screen inherits two things from whichever database it loads: the labels it trains on and the pool it ranks. Neither is a property of the method, and neither can be measured from inside a single database, because each is a property of the record as a whole. Measured on the resolved union, both are worse than the database a campaign loads is able to report. Where two independent sources describe the same crystal they disagree by more than the margin conventionally used to decide whether a compound is stable; and almost a quarter of the labelled record is unavailable from any single provider. This section establishes the two in that order.

Measuring the labels begins with how much of the record is one crystal counted several times, which no member source can answer and a formula-level join would answer wrongly in both directions. Of the 13,902,836 structures carrying at least one reference, 3,233,769 carry two or more (Fig. 2a), and the distribution falls steeply while reaching far: 2,395,350 crystals are described by exactly two databases, 568,507 by three, 2,416 by thirteen and two by 25. Redundancy is concentrated where the field has concentrated its effort. The largest single overlap holds 1,672,167 shared crystals, and the eight largest are all between general-purpose density functional theory repositories (Fig. 2b); the enumerated framework libraries sit outside this entirely, sharing almost nothing with the general-purpose repositories or with one another, so the blank rows of the overlap matrix record where effort has gone rather than a gap in the corpus. These are the numbers a federated query cannot report and a formula-level join would misstate in both directions, and they bound what any cross-database evaluation can claim: for the largest overlapping pairs, a model trained on

one repository and tested on another is tested on hundreds of thousands of crystals it has already seen, unless the split is taken over structures.

Redundancy is only interesting if the redundant copies can be compared, and they can be compared here only because every value kept the method that produced it. Every method string is collapsed to a level of theory, and two databases are held against one another only where they answer the same question: a semilocal value is compared with a semilocal value, a measurement with a measurement, and a pair that meets only across that boundary is not compared at all (Fig. 2d–f; Supplementary Note S7). The comparison is possible on a large sample: 1,360,573 crystals carry a band gap from two or more databases at one level of theory, 846,303 a formation energy, 376,794 a hull distance, 145,346 a magnetisation and 15,303 a bulk modulus. Concordance falls as the quantity becomes harder to compute. Within 50 meV, 82.1% of shared band gaps agree. Within 50 meV atom^{-1}, 65.3% of formation energies and 64.2% of hull distances do, and within 5 GPa, 44.3% of bulk moduli and 49.3% of shear moduli do.

The tail matters more than the median. The tenth of shared formation energies that agree least differ by more than 205 meV atom^{-1} (Fig. 2e), which is several times the hull-distance margins conventionally used to call a compound stable [27, 28]. Whether a candidate passes such a filter can therefore depend on which database supplied its energy, and in a corpus without structural identity that dependence is not merely unquantified but unobservable. This is the first of the section's two results.

It survives three checks, and each sharpens it rather than softening it. The first is that agreement between two databases is not evidence of independence, so the concordant fraction quoted above overstates it. Resolving the comparison by pair separates two regimes that a single agreement figure would merge (Fig. 2f). Some pairs agree almost exactly. Two of the largest repositories share 138,959 crystals whose band gaps have a median absolute difference of exactly zero and agree within tolerance for 92.3% of them. That is not independent confirmation but the signature of shared provenance, two datasets carrying one underlying calculation, and reading it as corroboration would invert its meaning. Other pairs disagree systematically. For formation energy the median signed difference between the Materials Project and JARVIS-DFT is +70.7 meV atom^{-1} against a median absolute difference of 92.7 meV atom^{-1}, so nearly all of the disagreement carries one sign. The corresponding figures are +171 against 177 meV atom^{-1} for the Materials Project against AFLOW. Pearson correlations across these pairs remain between 0.85 and 0.99, so what separates the databases is an offset and a scale rather than noise. This is precisely the structure that merging

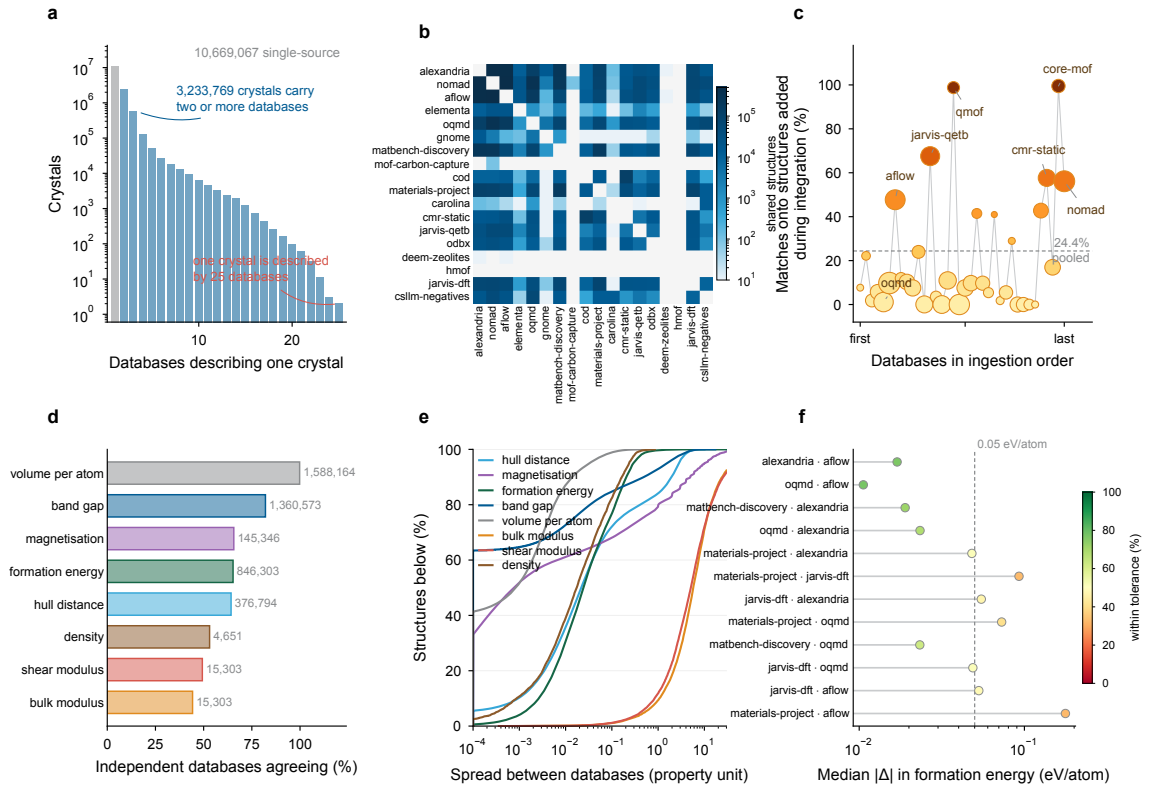


Fig. 2 Redundancy between databases, and whether the duplicates agree. a, Number of databases describing one crystal. Grey marks the crystals held by a single database, blue those held by two or more. **b**, Shared crystals between the databases annotating the most structures, on a logarithmic colour scale. An empty cell marks a pair with no crystal in common rather than missing data. **c**, The growing recognition substrate. For each database, in ingestion order, the fraction of its matches landing on a crystal absent from the starting pool and admitted during assembly. Marker area scales with the number of matches, and the dashed line is the value pooled over all databases. **d**, For each comparable property, the fraction of multi-source crystals whose independent values fall within the tolerance stated in Supplementary Note S7. The figure beside each bar is the number of crystals carrying that property from two or more databases at one level of theory. **e**, Cumulative distribution of the spread between databases, in the property's own unit for energies, magnetisation and moduli and as a relative difference for density and volume per atom. **f**, Median absolute difference in formation energy for the database pairs sharing the most crystals, coloured by the fraction agreeing within the tolerance marked by the dashed line.

columns by name would absorb without trace, and precisely the structure a model trained on the merged column would learn as if it were physics.

The second check is that the offsets are not a failure of the underlying theory, so they will not be removed by better software. Forty independent implementations of density functional theory, run on the same systems with the same functional, agree on the equation of state to well within the differences reported here [29]. What separates two databases is therefore not the solver but the accumulation of choices each provider makes around it, the pseudopotentials, the functional, the convergence criteria and the elemental or anionic corrections applied afterwards, none of which is visible in a column of numbers labelled formation energy. The reproducibility of the theory bounds what a careful user could hope for; the disagreement measured here is what a user actually receives.

The third check is that the disagreement is not an artefact of comparing different quantities, and the method strings are what make it checkable. A further 24,380 crystals carry a formation energy from two databases that worked at different levels of theory; they are deliberately left uncompared, because the difference between a tight-binding and a semilocal energy is not a disagreement. For density the excluded overlap is the larger part of it, 37,770 crystals against 4,651 compared, since the experimental catalogues and the computed repositories are not reporting the same quantity. Excluding these pairs lowers the measured disagreement rather than raising it, so the figures above are what remains after the comparison has been made fair. A union that merged the columns by name would have reported every excluded pair as a discrepancy instead.

Measuring the disagreement is not all the comparison is good for. It also identifies what no single-source check would question, at two scales. At the scale of the value, 1,161 of the shared elastic moduli lie outside any physically plausible range, being negative or above 1000 GPa, the largest reaching 1.4×10^{13} GPa. These are unconverged elastic-tensor fits published as they stand, and in isolation each is simply a number in a column; set against a partner value on the same crystal, each is immediately identifiable. At the scale of the column, one source that carries several levels of theory for the same crystal turns out to supply one band-gap column that tracks an independent repository closely and another, at a different level, that correlates with it at $r = 0.09$ because it reports almost every crystal as a metal (Supplementary Note S7). From inside that column the result reads as a list of metals and nothing marks it as wrong. It is legible here only because the same crystals carry an independent value, and separable from the usable column beside it only because the level of theory travelled with every number.

The pool is the second inheritance, and the question it raises is how much of it carries any label at all. The answer is: very little, and what there is was not distributed by importance, nor even by cost. Of the 166 properties in the registry, 151 carry at least one value. Treating the pool and those 151 properties as a matrix gives 2,203,163,235 cells, of which 99,493,466 are occupied, so the matrix is 4.52% full (Fig. 3a). Occupancy is extremely uneven: nine properties cover more than 40% of the pool, while 97 of the 151 cover less than 1% and 58 cover less than 0.1% (Supplementary Table S2). Half of all crystals carry at most seven of the 151 properties (Fig. 3c).

The gradient does not follow scientific importance. It follows price (Fig. 3b). Grouping 69 general-purpose properties, excluding those whose domain is one material family, by the calculation each requires (Supplementary Note S8), 19 are read off a single self-consistent calculation a database performs anyway. Their median coverage is 4.56% of the pool and eight exceed 10%. The remaining 50 require a calculation of their own, from an extra slab or strained cell through linear response to anharmonic phonons, transport integrals and hybrid functionals. Their median coverage is 0.107%, lower by a factor of 43, and *not one of the fifty reaches 10%.*

Cost is not the whole ordering, and the exception identifies what the ordering really is. Within the second group, linear-response properties are the more expensive to compute yet reach a higher median coverage (1.06%) than the properties obtainable from a few extra self-consistent runs (0.029%), a thirty-six-fold inversion of what price alone predicts. The elastic and dielectric tensors were the explicit target of dedicated high-throughput campaigns [30, 31], while quantities such as the work function, cheaper to obtain, were not the target of any. What the coverage gradient records is therefore not the difficulty of a calculation but whether somebody once organised a campaign to perform it, and coverage is a map of the field's attention rather than of physical cost. The distinction matters for what can be inferred from a gap: an expensive property that no campaign ever addressed is not evidence that the property is unimportant, only that its turn has not come. This map is legible only once the databases sit on one structure pool, because no single repository can report on effort it did not spend.

The same stratification governs non-scalar data and provenance. The curve store holds 1,198,897 spectra across ten types but reaches only 298,662 crystals, 2.0% of the pool (Fig. 3e). X-ray absorption spectra are the most numerous of the ten and among the least widely spread, because one crystal carries a separate spectrum for every absorbing element and edge, so the count of spectra held says little about the number of crystals reached. And only 66 of the 151 populated properties are reported by two or more databases, and the other 85 have a single supplier and therefore

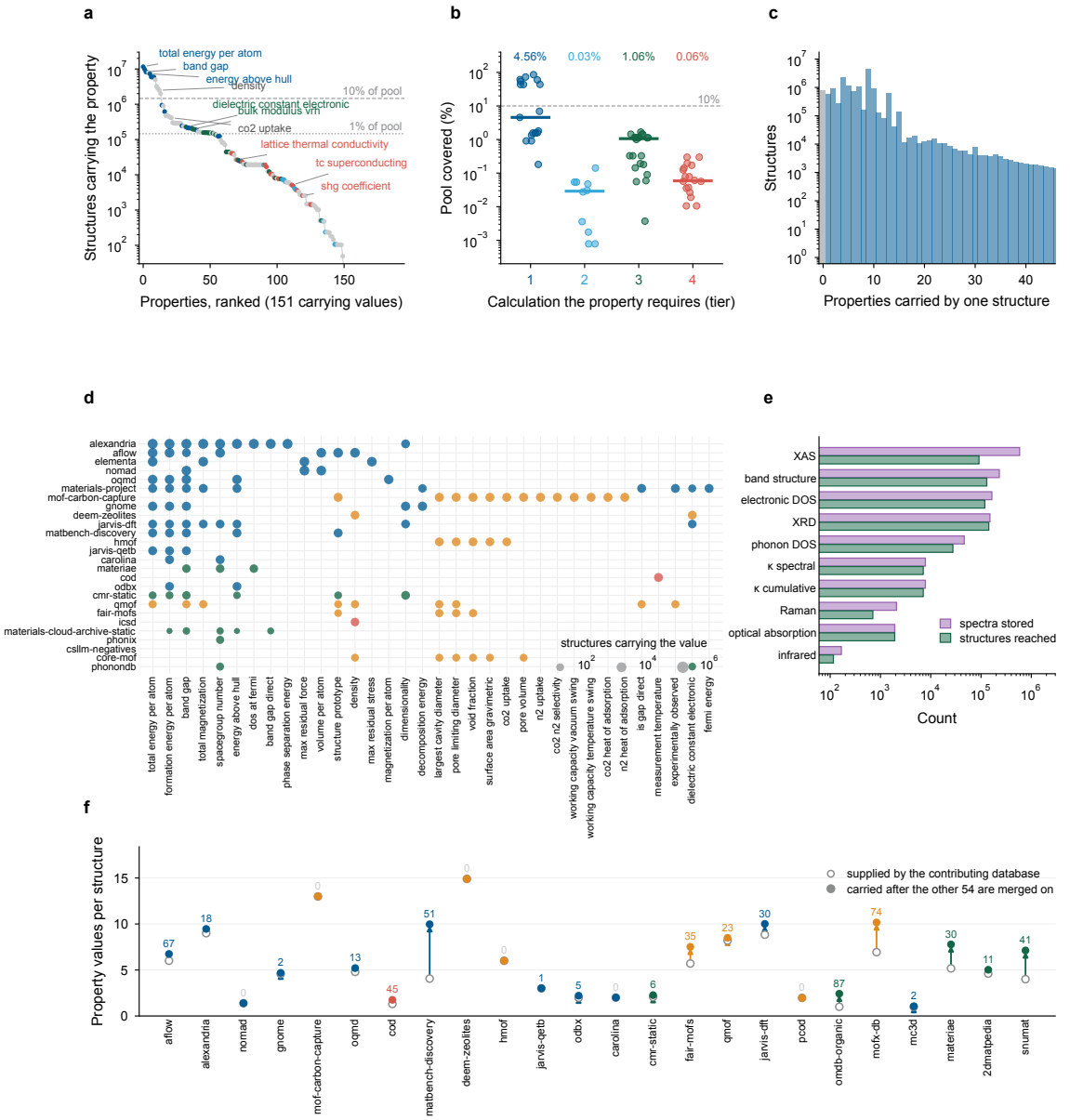


Fig. 3 What the corpus contains, and what it does not. a, Every property carrying at least one value, ranked by the number of crystals that carry it. Colour marks the calculation tier assigned in **b**. Grey marks properties outside that classification, whose domain is a single material family. The horizontal lines mark the stated fractions of the pool. **b**, Coverage of the general-purpose properties grouped by the calculation each requires: tier 1, readable from a single self-consistent run; tier 2, a few extra self-consistent runs such as a slab or a strained cell; tier 3, linear response or finite differences over many displacements; tier 4, anharmonic, transport or beyond-semilocal treatment. Each point is one property, horizontal bars are group medians and the figures above them give those medians. **c**, How many populated properties a single crystal carries. **d**, Which database supplies which property, for the most widely covered properties and the databases annotating the most crystals. Marker area scales with the number of crystals carrying that value, and colour marks the database family. **e**, The spectrum store by curve type: spectra held, and the number of distinct crystals they reach. **f**, What each database's own structures gained from the others. Every structure has one contributing database, the one that introduced it; the open marker is the property record that database supplied and the filled marker what the structure carries once the others are merged onto it, averaged over the structures each contributed. The figure above each database is the percentage of its structures that gained at least one value it never reported. The databases contributing the most structures are shown.

no independent check of any kind. Of the 99,493,466 occupied cells, 6.3% carry values from two or more sources, which is the exact extent of the corpus on which the comparison of the labels above is possible.

Sparse as the matrix is, part of it was filled by the integration itself, and measuring that part gives the section's second result (Fig. 3f). Every structure has exactly one contributor, the database that introduced it, and we ask how much of its property record came from somewhere else. Of the 6,266,517 structures admitted during assembly that carry any value at all, 1,637,460, or 26.1%, carry at least one their own contributor never reported; 3,204,768 of their 40,798,399 filled cells, 7.9%, were supplied by another database. The effect is largest where the chemistry is crowded. Two thirds of the structures AFLOW contributed gained a value it never published, as did half of those from the Matbench Discovery set and 45% of the crystallographic entries from COD, which arrive carrying little beyond their measured geometry and leave with a computed value attached to nearly half of them. It is smallest, close to zero, for the enumerated libraries of hypothetical zeolites and metal–organic frameworks, whose structures nobody else has computed: they give to the record without receiving from it.

This figure depends on ingestion order, since a database's own structures are by construction those no earlier source held, and only later arrivals can annotate them; the source ingested last is therefore necessarily at zero. The order-independent statement is that 3,153,748 crystals, 22.9% of all that carry any value, draw their property record from more than one database. Either way the same conclusion follows. Almost a quarter of the labelled record is not available from any single provider, so a campaign that loads one database ranks a smaller pool than the one that exists and has no indication of what it is missing. The two inheritances therefore bound a screen from opposite directions: the pool fixes what it can be asked about, and the labels fix how far its answer can be trusted.

2.3 Probing the limit of forward screening

Both bounds of Section 2.2 move together if the property layer is completed by prediction over the whole record rather than over one provider's slice of it. Whether such a completion succeeds is a question about trained models, and it is answered when their held-out numbers exist. What can be established now is that the problem is well posed on this corpus and is not well posed on any single database, and that is what this section does.

A completion needs three things no single database provides: structures deduplicated, so that a train–test split falls on distinct crystals rather than on records; labels that keep the quantity they denote rather than the column they were filed under; and, beside each label, the spread of the sources that reported it, which fixes the resolution at which a predicted value can be read, since a prediction closer to the truth than the sources are to each other cannot be shown to be so. The corpus supplies all three, and imposes four constraints on how they may be used. Filling the same gap by calculation would take an effort no single group can supply.

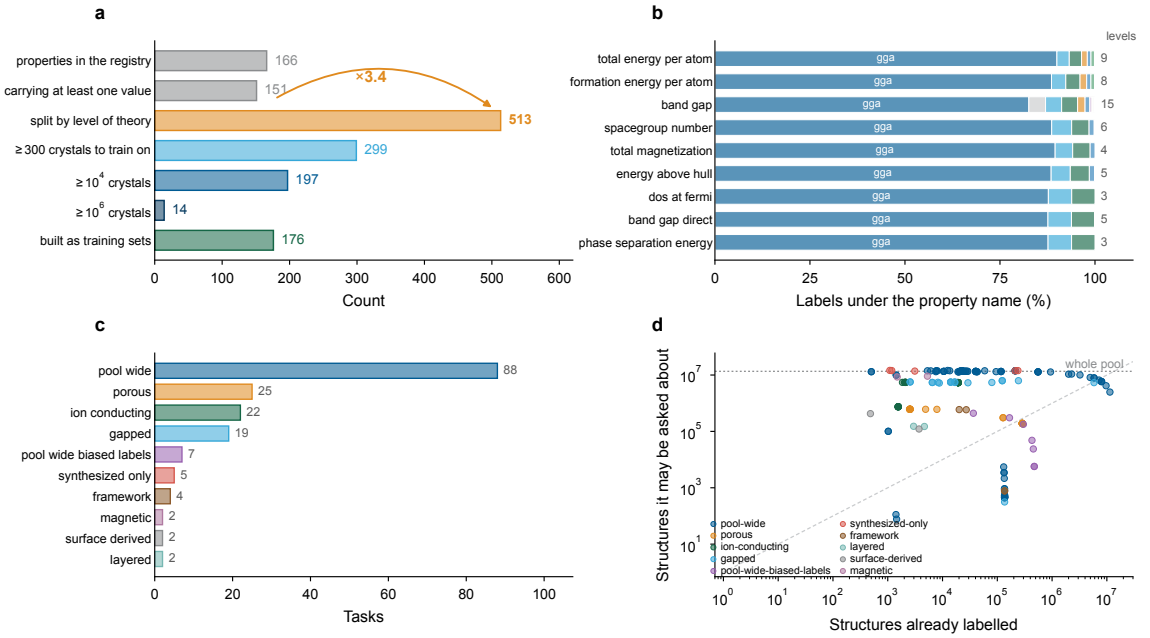


Fig. 4 What the corpus offers a learned completion of the property layer. a, How the registered properties become separable learning tasks and then realised training sets. The arrow marks the one stage that multiplies rather than filters, splitting a property name by level of theory. **b**, Composition of the most widely covered properties by level of theory, as a fraction of the labels filed under each property name. The figure at the right gives the number of distinct levels. **c**, The constructed training sets by declared applicability domain, the set of structures for which the property is defined at all. **d**, For each training set, the crystals already labelled against the crystals the task may legitimately be asked about once its domain is enforced. The dotted line marks the whole pool and the dashed diagonal marks parity. Panels reporting the trained models are added when their held-out numbers exist.

A census of the database establishes how much is actually trainable (Fig. 4a). The unit that matters is not the property but the property at one level of theory, because a name such as band gap covers several different physical quantities that a model cannot distinguish from structure alone. Split that way, the 151 populated properties yield 513 candidate tasks, set against the tens that existing property-prediction and stability benchmarks provide [32, 33]. Of these, 165 regression tasks have at least 10^4 crystals to train on, 100 have at least 10^5, and 13 have at least 10^6. Constructing the training sets under a default policy that keeps the largest level of theory per property produced

176 sets covering 116 properties and 93,864,594 labels, each with a chemistry-aware train, validation and test split.

The four constraints are each a direct consequence of the integration reported above. The first is that a property name is not a label definition (Fig. 4b). Band gap carries 15 distinct levels of theory, of which the semilocal group holds 83% and no other reaches 5%. The systematic offsets between them are of order 1 eV, which a model cannot infer from the structure and would instead learn as a proxy for which database supplied the row. The second is that some quantities are condition-dependent rather than multiply determined: carbon dioxide uptake is reported at several distinct grand-canonical conditions, so a material legitimately carries several different correct values and taking their median produces a number that is not any physical quantity. Neither the level of theory nor the condition is resolved away, and neither is the cross-source spread of Section 2.2: across the training sets, 5,856,861 crystals carry labels from two or more databases, and that spread is kept beside the resolved value so that it can weight the loss rather than being averaged into it.

The third is that the data cannot be split at random. The same compound recurs in the pool under different accession numbers, so a random split places identical chemistry in training and test, and the resulting scores are optimistic in exactly the regime that matters, since the crystals a completion is wanted for are the ones whose chemistry the training set never saw. Holding out whole chemical systems is the right default, but it degenerates on the enumerated libraries, where an entire task can occupy a single chemical system and a chemistry-level holdout leaves the validation and test folds empty. The split rule falls back for those tasks and records which level it used (Supplementary Note S9).

The fourth is that the completion has a domain (Fig. 4c,d). Carbon dioxide uptake is a grand-canonical adsorption quantity: it is not zero for a dense intermetallic, it is undefined, because the simulation cell has nowhere to insert a molecule. Of the 176 training sets, 63 draw more than 90% of their labels from a single material family (Fig. 4c). Enforcing their declared domains changes the size of the completion problem substantially: 109 of the 176 tasks are defined on less than half the pool, and the number of predictions that lie inside a declared domain is 1,105,593,143 rather than the 2.57 billion that asking every task about every crystal would produce, a factor of 2.3. The difference is not a loss of coverage. It is the removal of numbers that would have looked like data. Plotting what a task may be asked about against what it already knows separates the two situations that matter (Fig. 4d): a task lying far above the diagonal has the most to gain, because its domain

is large and its labels are few, while a task lying to the right and low is one whose domain is already nearly labelled and where a model has little left to add.

One dependency in that structure is worth stating because it closes the loop with Section 2.1. Twenty-one of the tasks are gated on a property the database does not yet hold: whether a crystal has a gap, or whether it is magnetically ordered, is unknown for 108,246,933 structure-task pairs, so their admissibility can only be decided once the corresponding tier-1 models have run. The completion therefore proceeds in stages, and each stage enlarges the domain over which the next one is defined, in the same way that each admitted structure enlarged the substrate against which the next database was matched.

What such a completion would settle is worth stating before it is attempted, because it is the reason the corpus was assembled rather than merely enlarged. Searching the known set for a material that already has the property wanted does not require that the property have been computed, because a surrogate model can rank crystals nobody has evaluated, and that is how such campaigns are run. What is bounded is the pool over which the ranking is taken. A campaign screens the repository it loaded, and Section 2.2 sets both what any single repository can offer as training signal and how far its labels can be trusted where a second source exists. Running the completion over the whole corpus, within the domains of Fig. 4c,d, changes both terms at once: the pool becomes the deduplicated record rather than one provider's slice of it, and every label carries the spread of the sources that reported it.

That is what measures a quantity the field has never had: how much of the known record has been passed over for want of a calculation rather than for want of merit. A large yield would mean that forward screening has been operating far inside its own reach. A small one would mean the reach is nearly exhausted and that further progress has to come from designing structures rather than finding them. The same corpus serves that second route as well, since a generative model needs exactly what it holds: structures deduplicated across the whole known record, so that novelty is judged against everything published rather than against one repository, and property labels that name the quantity they denote, so that a model can be conditioned on a defined target. Which of the two answers the completed layer gives is an empirical question, and it is the one the next stage is built to decide.

[Pending: the trained models]

The models are still training; no number below is in hand. This block becomes the closing passage of Section 2.3 and adds panels to Fig. 4: the multi-task backbone over the tier-1 tasks and the fine-tuned heads over the smaller ones, held-out accuracy per task set against the multi-source noise floor of Section 2.2, the coverage the completed layer reaches for the properties of Fig. 3b, and the boundary within which a predicted entry may be used. It is written when the held-out numbers exist, so that no claim precedes its evidence.

[Pending: what the completion nominates]

The candidate list does not exist yet. When the models of Section 2.3 have run, this block reports, for the properties whose held-out error falls below the multi-source spread of Section 2.2, how many crystals already in the record are nominated for a property no source has ever reported for them, how many of the highest-ranked survive recomputation, and how many survive a search of the literature the databases do not index. Selecting the extreme of a ranked prediction selects on prediction error as well as on the property, so the ranking is taken on predicted value gated by ensemble spread, and the selection effect is reported rather than assumed away.

2.4 Judging the novelty claimed by inverse design

The record that bounds forward screening is also the reference inverse design needs, and the point of this section is that a reported novelty fraction measures that reference at least as much as it measures the model. A generative model counts as new whatever is absent from the reference set it was given, so a structure missing from one repository but held by another is counted as new; the reference sets in current use are drawn from a handful of the largest repositories [19, 20], and the three assessments cited in the Introduction all turned on exactly that [21–23].

Two distinct failures sit behind those three assessments, and the corpus assembled here has already met both from the other direction. The first is a criterion too permissive, counting a known prototype under a new decoration as a new material. The second is a criterion that misses a match it should have made, which is what happens when an ordered approximant is generated for a parent phase that is disordered on one site. Section 2.1 reports the first as over-merging in the enumerated framework libraries, where a bonding-graph fingerprint collapses more than 10^5 distinct hypothetical frameworks onto a few hundred pool materials, and Supplementary Note S2 states the domain within which that fingerprint is valid. Both are failures of the criterion rather than of the generator, and no published novelty figure states which criterion produced it. What this record offers the route is therefore not a better score but a reference that can be named: the resolved corpus rather than

a few of its largest members, and a matching rule whose domain of validity is stated rather than assumed.

[Pending: novelty under a stated identity criterion]

Not yet run. The audit applies the matcher of Section 2.1 to the structure sets released by published generative campaigns, and reports the fraction the resolved record already contains under each of four identity criteria in turn: exact structure match, the structural fingerprint used here, that fingerprint extended to admit an ordered approximant of a disordered parent, and a prototype-level key. The point of the ladder is that a novelty fraction is a property of the criterion as much as of the model, and no published figure states which criterion produced it. The disorder-tolerant path is not implemented yet, and Supplementary Note S2 states why the present fingerprint would not recognise such a case.

3 Discussion

Holding the public record as one structurally resolved space changes what can be asked of it. A campaign that used to begin with account requests, bulk downloads and format conversion begins with a query. A crystal is one row rather than six, so counting is possible, and every value names the calculation behind it, so comparison is possible. More consequentially, a repository can report only on its own contents, whereas a resolved union reports on the field, and two of the things it reports were not what a user of any single source would assume. Independent sources computing the same formation energy for the same crystal at the same level of theory agree within 50 meV atom^{-1} barely two times in three, and because the discrepancy is a systematic offset rather than scatter it does not average away with sample size: a corpus that merged the contributing columns would hand a model the offset to learn as though it were physics. In the other direction, sources that agree to the last decimal are usually carrying one calculation twice, so agreement between repositories should be read as shared provenance until shown otherwise. Neither observation requires new computation. Both require only that the record be held in a form where the same crystal is the same row and every number remembers where it came from.

For forward screening, the record moves the limit from where it is usually placed. The route is not confined to materials somebody has already computed, because a surrogate model ranks the uncalculated as readily as the calculated. It is confined by the pool it is asked about and by the labels it was taught from. A campaign screens one repository's holdings and trains on labels whose providers disagree by the margins reported above, and both of those are properties of the record rather than of the method. Running the completion over the whole corpus, inside the domains where each property is defined, moves both at once, and answers a question the field has not been able

to pose: how much of the known record has been passed over for want of a calculation rather than for want of merit. Both answers are consequential. A large yield would mean forward screening has been operating far inside its reach, with already-synthesisable materials waiting to be recognised. A small one would mean the known set is close to exhausted and progress has to come from designing structures rather than finding them.

For inverse design, the record is the reference a claim of novelty needs, and the same record serves the route in two ways that are easy to conflate. It supplies what a generative model needs in order to work: deduplicated structures, and labels that name the quantity they denote, so that generation can be conditioned on a defined target rather than on a column heading. It also supplies what such a model needs in order to be believed. The failures reported against recent campaigns [21–23] were not failures to generate but failures to recognise what the record already contained, and a claim of novelty is only as strong as the reference set and the identity criterion behind it. On this reading the two routes are not competitors to be ranked. Forward screening is bounded by how much of the known set it is asked about, inverse design by how much of the known set its novelty is measured against, and both bounds are the same missing thing seen twice. Supplying that record is what makes the two routes comparable, and eventually complementary, rather than separate literatures.

What makes this maintainable is that admission and matching are the same mechanism seen from two sides. A record that matches nothing is not an error to be logged but the corpus's next member, and once admitted it becomes part of the criterion by which the next record is judged. In this corpus 24.4% of all matches resolved onto crystals that assembly itself had added, approaching unity for chemistries the starting pool did not cover, and the last source ingested, matched against the largest substrate, found more than half of what it recognised among them. The same relation reappears in the completion stage, where twenty-one tasks can be gated only once earlier models have supplied the property their domain is defined by. A space built this way has no final state, which is what a merged snapshot cannot offer as its sources keep moving. Three boundaries should be read with the results. Structural fingerprinting has a domain of validity that enumerated framework libraries fall outside, and the composite key that rescues them is a remedy for a known failure rather than a general solution. Agreement is measurable only where two providers report the same quantity under comparable methods, so the disagreement reported here is measured on the part of the corpus most likely to be well behaved. And no value was recomputed: every number is as its source reported it, and the union inherits whatever errors those sources contain. What it adds is the

ability to see when two of them disagree. The full set of limitations, including the licence boundary and the structural rather than geometric applicability gate, is Supplementary Note S10.

4 Methods

4.1 Sources and their ingestion

Each of the 55 databases is reached through one adapter emitting a stream of records, each carrying the source's native identifier, its property values with units and methods, its structure where the licence permits redistribution, and any cross-database identifiers the source is authoritative for. Sources that may not redistribute structures are marked properties-only in code, so no structure of theirs can enter the pool. The complete inventory, with each database's licence, contribution and reference routes, is Supplementary Table S1. The adapter protocol and staging pipeline are Supplementary Note S1.

Property names are mapped onto a canonical registry of 166 entries, each with a fixed unit and value kind. A source field enters the registry under an existing name only when the quantity is the same physical quantity, and otherwise gains its own name. Method strings are required to distinguish values that a source reports for the same material under different settings, and the loader treats (native identifier, property, method) as the uniqueness key. The registry and its boundary cases are Supplementary Note S3.

4.2 Identity resolution

Structures are fingerprinted with the BAWL hasher [19] under the protocol used for the original deduplicated merge, so that fingerprints computed here are comparable with those already in the pool: parse, reject cells above 500 sites and disordered occupancies, reduce to the primitive cell with pymatgen [34], reject primitive cells above 300 sites, then hash.

Matching proceeds in two ordered paths against a map holding fingerprint-to-material and (source, identifier)-to-material entries. A record is looked up first by every authoritative cross-database identifier it carries, and then by fingerprint. Chemical formula is never a match criterion. A record matching nothing is admitted as a new pool material, and its fingerprint and identifiers enter the map, which is the mechanism measured in Section 2.1. Enumerated framework libraries, whose siblings are indistinguishable to a bonding-graph fingerprint, are detected at the level of the source and admitted under a composite key adding the cell volume per site (Supplementary Note S2).

Every match records the material, source, native identifier and route taken. Values are never overwritten or averaged: several databases reporting one property for one material coexist as separate rows.

4.3 Audit

Completeness is verified by three checks applied to each source independently: the row count the loader reported must equal the row count the database holds, every staged row must be either loaded or accounted for as an identifier no pool material corresponds to, and every property name present in the staged data must be present in the database for that source. The third catches a field the adapter maps but the loader never lands, which leaves row totals plausible while an entire quantity goes missing. The completed database passes all three for all 55 sources (Supplementary Note S4).

Every aggregate statistic reported here is computed with a single grouped scan over the property table. The natural alternative, a correlated subquery per source, rescans 132 million rows 55 times and evaluates each source against a different snapshot.

4.4 Cross-database comparison

Agreement was computed for eight properties whose values are comparable between databases once the level of theory is fixed, whereas extensive and convention-dependent quantities were excluded, because two databases disagreeing on a total energy reflects a different reference state rather than a different answer. Each method string is collapsed to a theory level and then to one of four comparable sets, and values are compared only within a set, so that a source holding several levels for one crystal is not measured against its own mixture and a crystal whose only overlap crosses a set boundary is not compared at all. Within each material and set, each source was reduced to the median of its own values, so a source reporting several method strings of one level is counted once, and the spread was taken as the range across sources. Tolerances, the excluded quantities and the full pair table are Supplementary Note S7 and Table S4.

The 69 general-purpose properties of Fig. 3b were assigned by hand to four tiers by the calculation each requires, excluding properties whose domain is one material family. The assignment is listed in full in Supplementary Table S3 so that it can be disputed item by item.

4.5 Census and training-set construction

A task is defined as one property at one level of theory, since method strings collapsed onto a single name denote different physical quantities. Under that definition the populated properties yield 513 candidate tasks, and a default policy keeping the largest level of theory per property realises 176 of them as training sets over 93,864,594 labels. Condition-dependent properties are split by condition rather than reduced to a median, cross-database spread is retained beside the resolved label, and splits are taken over chemical systems, falling back through formula to random where a task occupies too few systems to yield non-empty folds. Each task carries a declared applicability domain and an empirical one measured from its training set, and predictions are admitted only inside both. The full protocol, the four constraints it addresses and the domain assignments are Supplementary Note S9.

4.6 Software and reproducibility

The pipeline is implemented in Python. Structure handling uses pymatgen [34] and symmetry analysis uses spglib [35]. The database is one SQLite file with two side-car binary blobs holding structures and spectra. Every figure is produced by a script reading comma-separated data files, and each of those files is produced by a named extraction script reading the published database read-only. No number quoted in this work was transcribed by hand.

Acknowledgements. We thank the maintainers of the 55 source databases integrated here. This work redistributes none of their data beyond what their licences permit, and depends entirely on their decision to publish in the first place.

Declarations

Competing interests. The authors declare no competing interests.

Data availability. The unified database is published as a versioned dataset on the Hugging Face Hub together with a development slice sized for interactive work. Five of the 55 sources are withheld from it: AFLOW's terms are academic and non-commercial, ELEMENTA and GNoME are CC BY-NC 4.0, Inorganic Crystal Structure Database records may be held only by a subscriber to the FIZ Karlsruhe licence, and NOMAD holds author-deposited calculations whose licence is set per upload rather than for the repository, so its values are withheld until the terms of each can be resolved. All five are searchable in a database built locally from the published pipeline and are filtered out

before any public release. The database contains no ICSD structure. The comma-separated data files behind every figure are provided as source data.

Code availability. The acquisition adapters, the fingerprinting and matching pipeline, the loader, the completeness audit, the extraction scripts and the figure scripts are available in the repository accompanying this work.

References

[1] Jain, A. *et al.* Commentary: The Materials Project: a materials genome approach to accelerating materials innovation. *APL Materials* **1**, 011002 (2013).

[2] Saal, J. E., Kirklin, S., Aykol, M., Meredig, B. & Wolverton, C. Materials design and discovery with high-throughput density functional theory: the Open Quantum Materials Database (OQMD). *JOM* **65**, 1501–1509 (2013).

[3] Curtarolo, S. *et al.* AFLOW: an automatic framework for high-throughput materials discovery. *Computational Materials Science* **58**, 218–226 (2012).

[4] Schmidt, J., Wang, H.-C., Cerqueira, T. F. T., Botti, S. & Marques, M. A. L. A dataset of 175k stable and metastable materials calculated with the PBEsol and SCAN functionals. *Scientific Data* **9**, 64 (2022).

[5] Kirklin, S. *et al.* The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies. *npj Computational Materials* **1**, 15010 (2015).

[6] Choudhary, K., Garrity, K. F., Reid, A. C. E. *et al.* The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design. *npj Computational Materials* **6**, 173 (2020).

[7] Draxl, C. & Scheffler, M. The NOMAD laboratory: from data sharing to artificial intelligence. *Journal of Physics: Materials* **2**, 036001 (2019).

[8] Talirz, L., Kumbhar, S., Passaro, E. *et al.* Materials Cloud, a platform for open computational science. *Scientific Data* **7**, 299 (2020).

[9] Xie, T., Fu, X., Ganea, O.-E., Barzilay, R. & Jaakkola, T. *Crystal diffusion variational autoencoder for periodic material generation. International Conference on Learning Representations (ICLR)* (2022). arXiv:2110.06197.

[10] Zeni, C. *et al.* A generative model for inorganic materials design. *Nature* **639**, 624–632 (2025).

[11] Merchant, A. *et al.* Scaling deep learning for materials discovery. *Nature* **624**, 80–85 (2023).

[12] Chen, C. & Ong, S. P. A universal graph deep learning interatomic potential for the periodic table. *Nature Computational Science* **2**, 718–728 (2022).

[13] Deng, B. *et al.* CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. *Nature Machine Intelligence* **5**, 1031–1041 (2023).

[14] Batatia, I. *et al.* A foundation model for atomistic materials chemistry (2023). The MACE-MP-0 foundation model, arXiv:2401.00096.

[15] Yang, H. *et al.* MatterSim: a deep learning atomistic model across elements, temperatures and pressures (2024). arXiv:2405.04967.

[16] Wood, B. M. *et al.* UMA: a family of universal models for atoms (2025). arXiv:2506.23971.

[17] Andersen, C. W., Armiento, R., Blokhin, E. *et al.* OPTIMADE, an API for exchanging materials data. *Scientific Data* **8**, 217 (2021).

[18] Evans, M. L., Bergsma, J., Merkys, A. *et al.* Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange. *Digital Discovery* **3**, 1509–1533 (2024).

[19] Ramlaoui, A. *et al.* LeMat-Traj: a scalable and unified dataset of materials trajectories for atomistic modeling (2025). arXiv:2508.20875.

[20] Betala, S. *et al.* LeMat-GenBench: a unified evaluation framework for crystal generative models (2025). arXiv:2512.04562.

[21] Cheetham, A. K. & Seshadri, R. Artificial intelligence driving materials discovery? Perspective on the article: Scaling deep learning for materials discovery. *Chemistry of Materials* **36**, 3490–3495 (2024).

[22] Juelsholt, M. Continued challenges in high-throughput materials predictions: MatterGen predicts compounds from the training dataset. *Materials Horizons* (2026). AUTHOR LIST AND PAGE RANGE NEED VERIFICATION against the published version; preprint doi 10.26434/chemrxiv-2025-mkls8.

[23] Szymanski, N. J. & Bartel, C. J. Establishing baselines for generative discovery of inorganic crystals. *Materials Horizons* (2025). VOLUME, PAGES AND DOI NEED VERIFICATION.

[24] Haastrup, S., Strange, M., Pandey, M. *et al.* The Computational 2D Materials Database: high-throughput modeling and discovery of atomically thin crystals. *2D Materials* **5**, 042002 (2018).

[25] Zhou, J., Shen, L., Costa, M. D. *et al.* 2DMatPedia, an open computational database of two-dimensional materials from top-down and bottom-up approaches. *Scientific Data* **6**, 86 (2019).

[26] Zhang, T. *et al.* Catalogue of topological electronic materials. *Nature* **566**, 475–479 (2019).

[27] Sun, W. *et al.* The thermodynamic scale of inorganic crystalline metastability. *Science Advances* **2**, e1600225 (2016).

[28] Bartel, C. J. *et al.* A critical examination of compound stability predictions from machine-learned formation energies. *npj Computational Materials* **6**, 97 (2020).

[29] Lejaeghere, K. *et al.* Reproducibility in density functional theory calculations of solids. *Science* **351**, aad3000 (2016).

[30] de Jong, M., Chen, W., Angsten, T. *et al.* Charting the complete elastic properties of inorganic crystalline compounds. *Scientific Data* **2**, 150009 (2015).

[31] Petousis, I., Mrdjenovich, D., Ballouz, E. *et al.* High-throughput screening of inorganic compounds for the discovery of novel dielectric and optical materials. *Scientific Data* **4**, 160134 (2017).

[32] Dunn, A., Wang, Q., Ganose, A., Dopp, D. & Jain, A. Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. *npj Computational Materials* **6**, 138 (2020).

[33] Riebesell, J. *et al.* A framework to evaluate machine learning crystal stability predictions. *Nature Machine Intelligence* **7**, 836–847 (2025).

[34] Ong, S. P. *et al.* Python Materials Genomics (pymatgen): a robust, open-source Python library for materials analysis. *Computational Materials Science* **68**, 314–319 (2013).

[35] Togo, A., Shinohara, K. & Tanaka, I. Spglib: a software library for crystal symmetry search. *Science and Technology of Advanced Materials: Methods* **4**, 2384822 (2024).